

Jess Fetter

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EDUCATION

Purdue University, West Lafayette

Expected 2028

Bachelor of Science in Mechanical Engineering, Management Minor

GPA: 4.0/4.0

Relevant Coursework: Calc 1-3, Linear Algebra, Differential Equations, Statics, Dynamics, Thermodynamics, Python for Data Science, Graphical Communication and Spatial Analysis (CAD)

PROFESSIONAL EXPERIENCE

Summer Engineering Intern

Summer 2026

Upper Story, Twin Cities, MN

- Refined a rough 3D toy car model and converted it into a manufacturable multi-part assembly using Rhino Sub-D and Autodesk Inventor.
- Engineered straight-pull, injection-moldable components by planning part splits, eliminating undercuts, and maintaining uniform wall thickness.
- Designed snap-fits and alignment features and iterated through multiple design cycles to improve fit, assembly speed, and part robustness.

Peer Teacher

Fall 2026- Present

Purdue University First Year Engineering, West Lafayette

- Guided student teams through engineering design challenges, helping them utilize the use of Python, MATLAB, and Excel.
- Supported grading and instructional coordination through weekly team meetings.

Lifeguard

Fall 2025 - Present

Purdue University Aquatics, West Lafayette

- Ensured patron safety and maintained facility readiness during recreational swim hours.

PROJECTS

CNC-Machined Chess Set

Spring 2026 - Present

- Machining a full chess set using CNC lathe and mill operations; generated all Fusion 360 CAM toolpaths for aluminum pawns and rooks.
- Built a contrasting hardwood checkerboard using traditional hand-tool woodworking techniques.

Amphibious Drone(Vertically Integrated Projects)

Fall 2025- Present

- Served as project manager, employed Gantt scheduling and weekly coordination across mechanical, CFD, and test-rig development.
- Designed a fully parametric propeller model with adjustable geometry for CFD validation and integration into a custom test rig.

Tapestry Crochet Project

Fall 2025

- Developed a Python program that converts images into simplified, stitch-accurate crochet patterns using K-means color clustering and multiple interpolation methods.
- Generated a complete PDF pattern through custom functions for color labeling, zig-zag traversal, palette mapping, and pattern formatting.

4 X 8 Hobbyist CNC

Summer 2024

- Built and assembled a 4x8 CNC based on the Lowrider L3 design; machined MDF struts using the partially completed machine.

EXTRACURRICULAR ACTIVITIES

Embedded Systems @ Purdue (ES@P)

Spring 2026 - Present

- Contributed mechanical design for a large-scale Pepper's Ghost illusion using an OLED display; developed structural and alignment features for stable integration.

Gamma Phi Beta Sorority, Risk Assistant

Fall 2025 - Present

- Selected through formal recruitment; participate in chapter operations, leadership development, and community engagement.

Women in Engineering

Fall 2025 - Present

- Engaged in monthly professional-development sessions; selected as a WIE mentor supporting first-year student transition and success.

SKILLS

CAD Software: Inventor, Fusion, Rhino, AutoCAD, Onshape, NX

Programming languages: Python, Java, MATLAB

Manufacturing & Fabrication: DFM, CNC Mill, CNC Lathe, Resin (SLA) 3D Printing, Soldering, CAM Toolpath Generation, Woodworking